TASK 3

API SECURITY RISK ANALYSIS

REPORT BY
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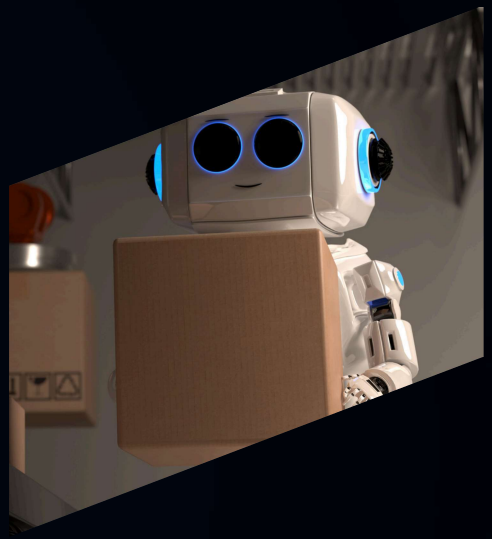
EXECUTIVE SUMMARY

APIs (Application Programming Interfaces) play an important role in modern applications by allowing different systems to communicate and exchange data.

However, insecure APIs can expose sensitive information and create security vulnerabilities.

The purpose of this API Security Risk Analysis is to identify potential security risks, analyze authentication and authorization mechanisms, and recommend improvements to strengthen API security.

TOOLS USED



The following tools were used during the API security analysis:

**Postman / Insomnia – To send API requests and test responses
**Browser Developer Tools – To inspect API calls and network traffic
**Sample Public APIs – For testing API functionality
**Google Docs / MS Word – For documenting findings

METHODOLOGY

The API security analysis was performed using the following steps:

** API Endpoint Identification

**Authorization Analysis

**Input Validation Testing

**Limiting Verification

IDENTIFIED SECURITY RISK

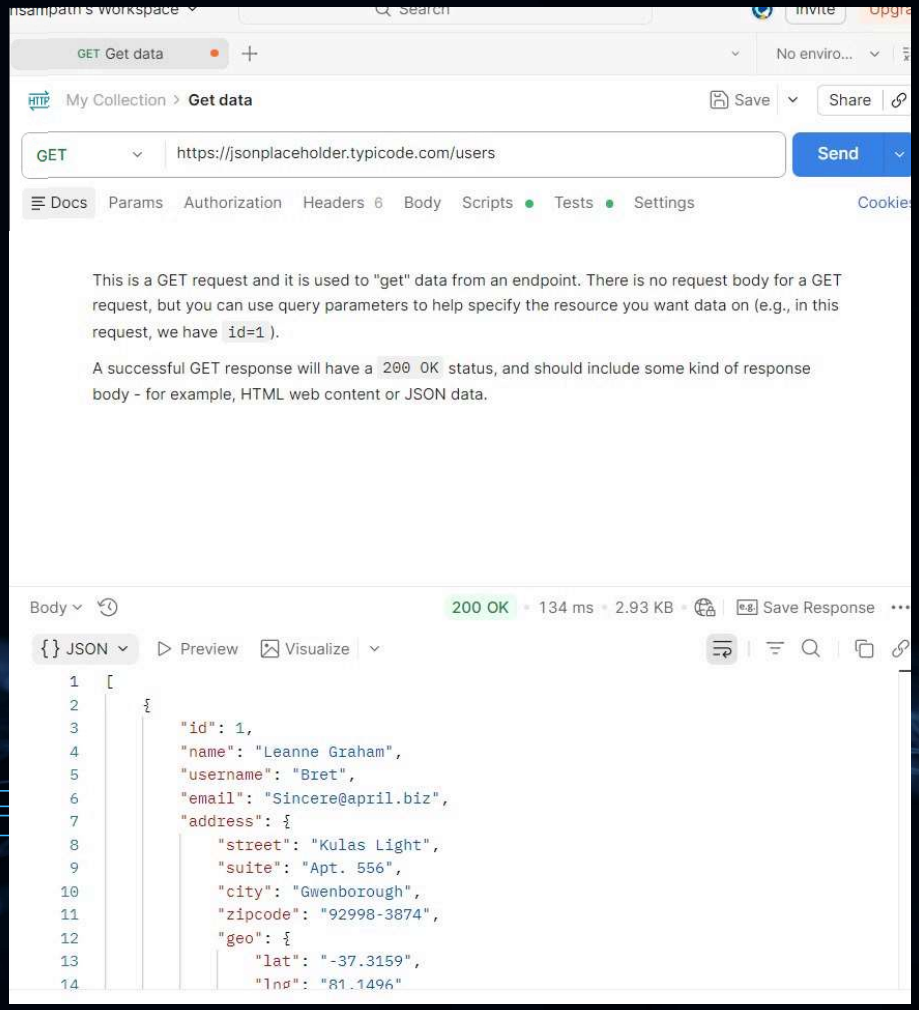
RISK 1 : MISSING AUTHENTICATION MECHANISM

Description:
The API allows public access to user data without requiring authentication

Impact:
Attackers can collect user information and misuse it.

Severity: Medium

Recommendation:
Implement token-based authentication (JWT or API keys).



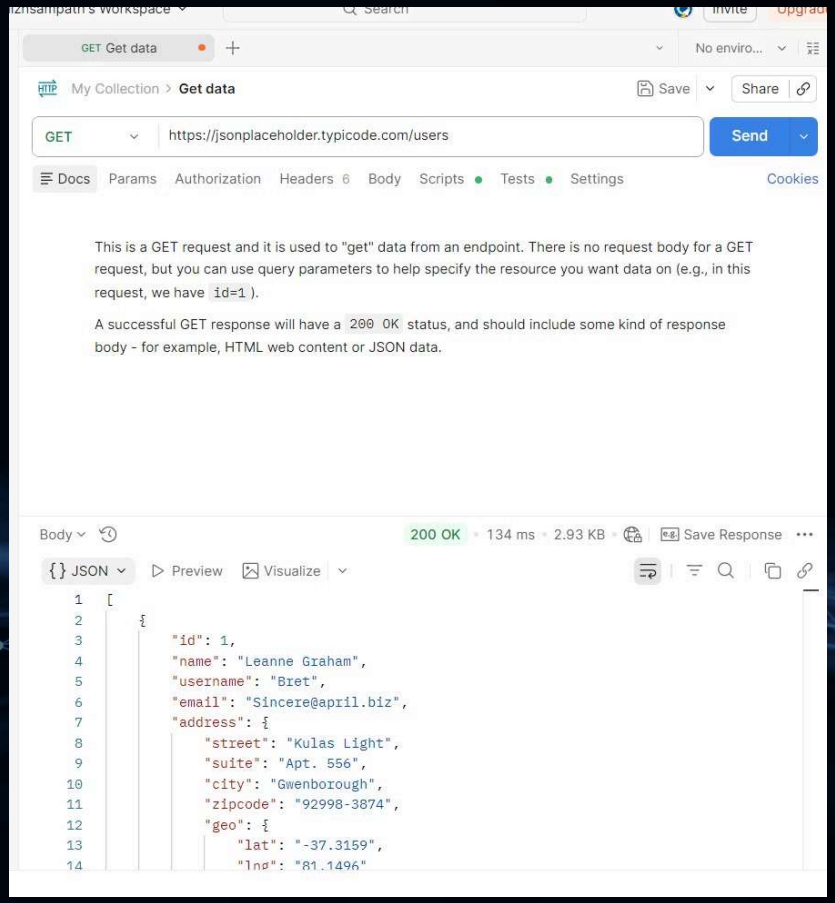
RISK 2 EXECUTIVE DATA EXPOSURE

Description:
The API exposes detailed user information including email and address.

Impact:
Sensitive data could be harvested.

Severity: Medium

Recommendation:
Return only necessary fields.



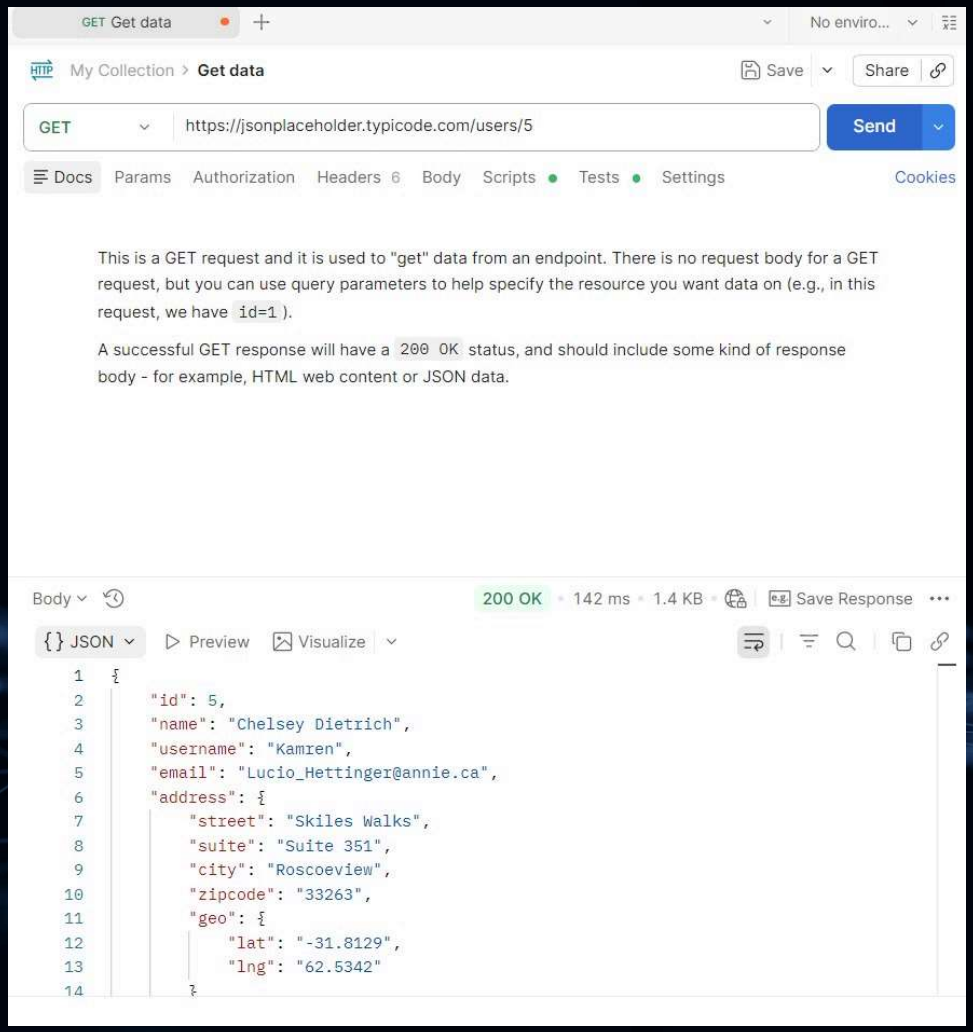
RISK 3: MISSING RATE LIMITING

Description:
No restrictions on repeated API requests.

Impact:
Brute-force or DoS attacks possible.

Severity: Medium

Recommendation:
Implement rate limiting (e.g., 100 requests/min



RISK 4: LACK OF INPUT VALIDATION

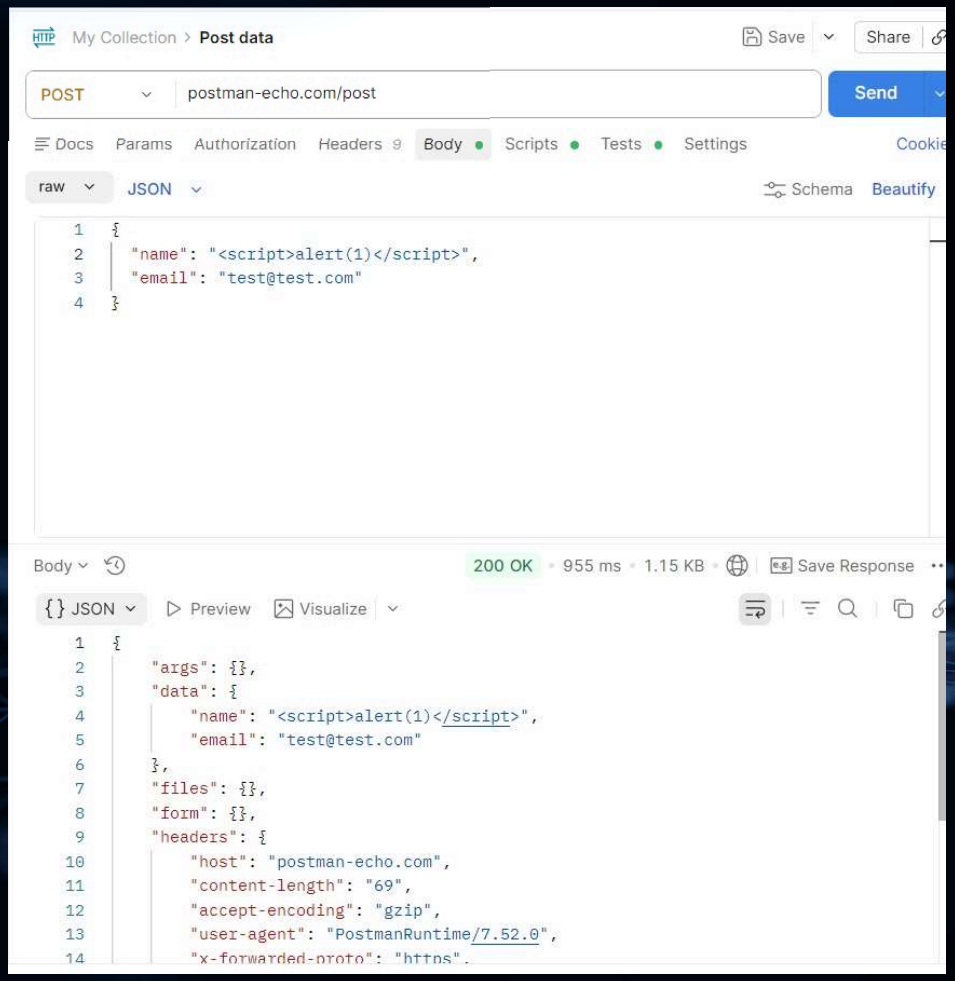
A POST request was sent with manipulated input:

{
"name": "<script>alert(1)</script>",
"email": "test@test.com"
}

The API accepted the input without validation.

Impact

Potential Cross-Site Scripting (XSS) if



Best Practices for API Security

To improve API security, the following best practices should be implemented:

Use strong authentication mechanisms
Implement proper authorization controls

Use HTTPS to encrypt API communication
Validate all user inputs

Apply rate limiting and request throttling
Log and monitor API activities
Regularly test APIs for vulnerabilities

CONCLUSION

APIs are an essential component of modern applications but can introduce security risks if not properly protected. This analysis identified several potential vulnerabilities such as insecure endpoints, weak authentication, data exposure, and missing rate limiting.